Memo Ozdincer 1010879319

Jeffery Tian 1010869945

Jadon Tsai 1010870283

November 12 3:00-6:00 PM

Applying the First Law of Thermodynamics and Ideal Gas Law to Determine Heat Loss and Work Done on a Control Volume of Air

1 Abstract

In this series of experiments, the first law of thermodynamics was applied to a mass of air inside a closed tank, heated with heaters and a propeller. The first law of thermodynamics states that energy can neither be created nor destroyed, only converted between forms. This law is used as a basis behind the design of virtually all thermal and mechanical systems, ranging from HVAC systems(Howell, 2017) to ironmaking (Sohn & Olivas-Martinez, 2014). The mass of air in the tank was initially determined using the ideal gas law. Although real gases deviate from ideal behaviour, and therefore the ideal gas law is not technically correct, the ideal gas law is a good enough model for most practical applications, particularly at moderate temperatures and pressures (Smith et al., 2018). These laws are the basis for more advanced equations used in engineering design and calculations. In the second part of the experiment, the heat loss from the tank walls was determined from taking the electrical power required to maintain a constant air temperature. We found that heat loss increased with both pressure and temperature. The average c_v value was roughly 93 kJ/(kg K), with an uncertainty larger than the value. We also determined that the work provided by the fan in the system was only 0.5% of the total work, and thus could be safely neglected.

2 Introduction

The first law of thermodynamics states that the change in energy ΔE in the system is equal to the work done on the system (W) plus the heat added (ΔQ) (all values in Joules (J)) (Smith et al., 2018).

$$\Delta E = W + \Delta Q$$

We assume that the gas in the tank behaves ideally, which allows us to use the ideal gas law:

$$PV = mRT$$

where P is pressure in Pascals (Pa), V is volume (m^3), m is the mass of the gas (kg), T is the absolute temperature in Kelvin (K), and R is the specific gas constant ($\frac{J}{kg*K}$).

The first objective of the experiment was to determine the rate of heat loss of the air in the tank at different pressures and temperatures. The second objective was to determine the specific heat capacity of air at constant volume (C_V)). The third objective was to determine the work done on the system by the propeller.

3 Method

3.1 Apparatus

The lab setup (Figure 1) consisted of two connected air tanks, connected via a system of valves, importantly a solenoid and micrometer needle valve, and a mass flow controller operated with LabVIEW, a software system used to facilitate data collection and processing. Pressure and temperature were measured with an integrated manometer and thermocouples respectively. A mass flow controller also logged the flow rate of air entering the system.

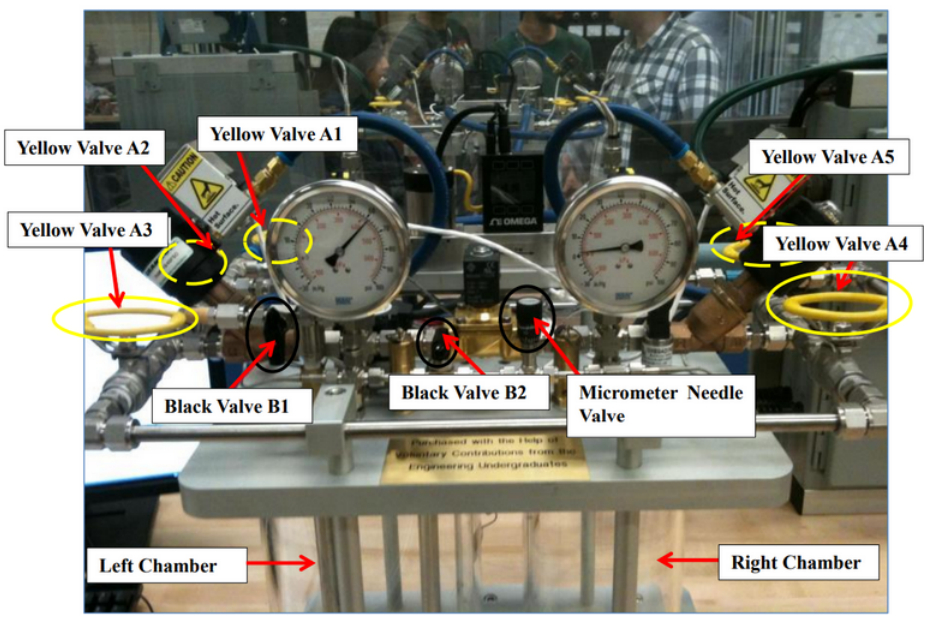


Figure 1: A picture of the experimental setup, with important valves labeled. (CHE260, 2025a)

3.2 Procedure

The following procedures were all taken from the lab manual (CHE260, 2025b) and performed as written in the manual, except for the temperature of the fourth trial, for which air was added until the pressure

reached 80 psi for consistency.

3.2.1 Procedure for Part 1

In Part 1, we wait for the left tank to reach equilibrium, while recording the mass flow rate and the final pressure and temperature of the gas. Note that in Figure 1, the left tank and right tank are denoted by left chamber and right chamber, respectively.

1. Record the ambient pressure and temperature using the manometer and thermometer in the lab.
2. Begin data collection by pressing "Start Collecting Data".
3. Set the pressure in the left tank to 40 psi, ensuring that the airflow is increased incrementally to avoid damage to the system.
4. Wait for the pressure and temperature to stabilize, stop recording and save the data to a file.
5. Press 'Reset' and proceed with the specific procedure associated with Part 2 in subsection 2.2.2.

3.2.2 Procedure for Part 2

In Part 2, we use two PID-controlled heaters to heat the left tank (with gas still inside) to a desired temperature, keep the gas at that temperature for a short duration, and run air through the tank to cool it back down again.

1. Begin data collection by pressing "Start Collecting Data".
2. Set the temperature to the desired temperature (as specified by the table given in the lab handout), and turn on the heaters by pressing the 'Hold' button.
3. Wait until the temperature reaches the specified temperature, then wait further for 5-10 minutes.
4. Stop data collection and turn off the heaters.
5. Run air through the left tank to allow the left tank to cool down. To do this:
   - Click "Start Collecting Data" to track the temperature, then open the Left Solenoid valve.
   - Open the Bar valve behind the left tank (we should probably have something for this in the picture), then wait for the temperature in the left tank to cool to the desired temperature.
   - Once the temperature in the tank has cooled to the desired temperature, close the Left Solenoid valve and wait for the temperature to stabilize. If the stabilized temperature is higher than desired, open the Left Solenoid valve and allow the tank to cool further.

6. Once the temperature has stabilized at the desired temperature, close the Bar valve.

7. Evacuate the tank by opening valve B1, then close valve B1.

8. Repeat steps 1-7 for the other pressures and temperatures.

4 Results and Discussion

In calculating results, we made a few assumptions. We assumed a rigid tank in order to have no boundary work, which is a pretty fair assumption as the tank size was fixed. We assumed uniform temperature across the tank. We additionally assumed ideal gas behaviour. We took the uncertainty in pressure to be 1 counts of the least significant digit, or 0.1 kPa. We took the uncertainty in energy to be 1 count of the least significant digit, or 0.1 kJ. We took the uncertainty in temperature to be ±1 K due to the fluctuations in final temperature. For brevity, we attach the temperature vs time and energy added over time graphs for the first trial. Other data is included in the appendix.

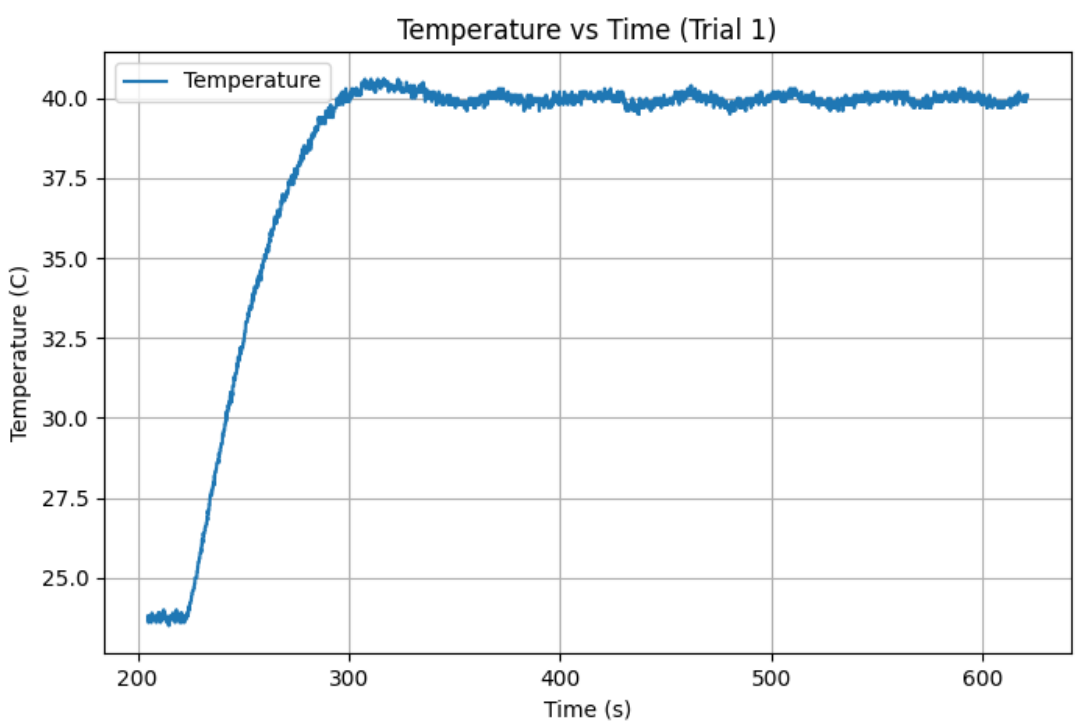


Figure 2: The temperature compared to the time. This graph was used to determine when the temperature reached the fixed value. Due to the oscillatory nature of the final temperature value, we take an uncertainty of ±1K

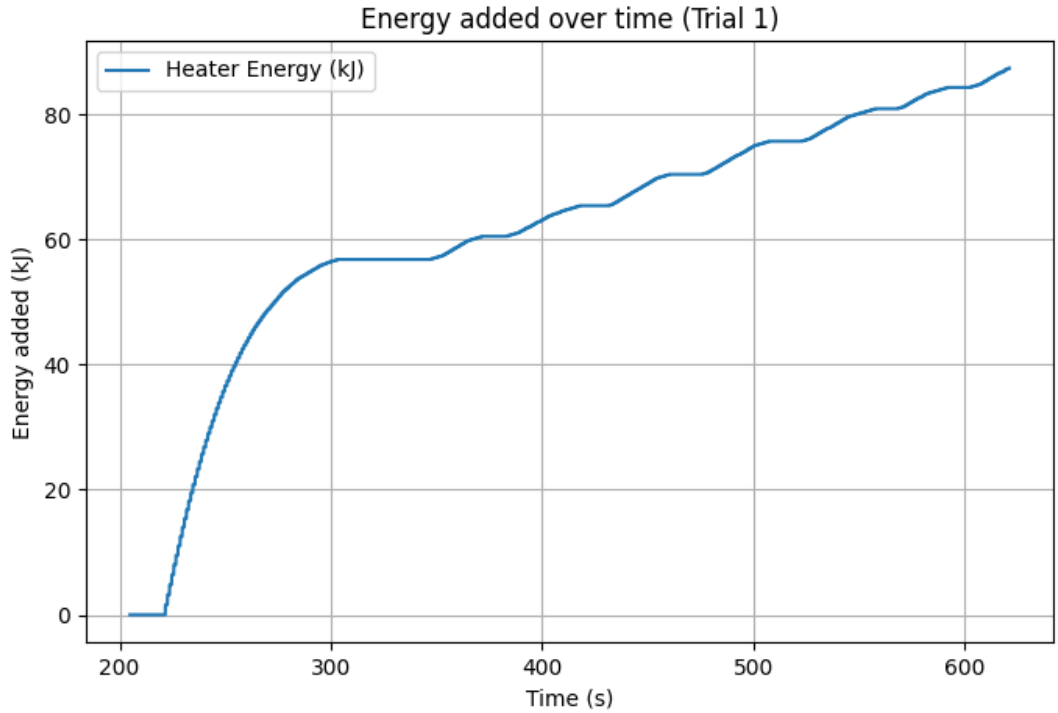


Figure 3: The energy added over time for the first trial. The heat loss was calculated by performing numerical integration over the time the tank is at steady temperature.

4.1 Determining the Mass in the Left Tank

To determine the specific heat capacity of air using the left tank, it is necessary to know the mass of air present within the left tank. To do this, we used the data collected by following the procedure in subsection 2.2.1 and the equation from (CHE260, 2025a):

$$m_{lt} = m_{at}\left[1 + \frac{1}{(\frac{P_2 T_1}{P_1 T_2}) - 1}\right]$$

Using this formula and the data taken in-lab, we determined the mass in the left tank for each trial. We propogated uncertainties as follows (example for trial 1):

$$\Delta m = m\left(\frac{\Delta P}{P} + \frac{\Delta T}{T}\right) = 0.03401\left(\frac{0.1}{367.4} + \frac{1}{313}\right) = 1.18 \times 10^{-4}\text{ kg}.$$

Thus, we take the uncertainty in mass to be ± 0.0001.

Table 1: A table showing the initial thermodynamic properties and the mass of the gas in the Left Tank.

Trial #	Final Pressure (kPa)	Final Temperature (K)	Mass of Gas Added (kg)	Total Mass of Gas (kg)
1	367.4 ±0.1	313 ± 1	0.0237 ±0.0001	0.0340 ±0.0001
2	642.5 ±0.1	313 ± 1	0.0237 ±0.0001	0.0341 ±0.0001
3	370.8 ±0.1	333 ± 1	0.0228 ±0.0001	0.0322±0.0001
4	642.5 ±0.1	333 ± 1	0.0479 ±0.0001	0.0579 ±0.0001

4.2 Determining Heat Loss and Specific Heat Capacity

To determine the specific heat capacity at constant volume c_v, we use the equation:

$$\Delta U = mc_v \Delta T$$

Where ΔU is the change in internal energy, m is the mass of the gas and ΔT is the change in temperature. Since boundary work can be taken to be negligible (we will show this in Part 3), $\Delta U \approx \Delta Q$, and we can rewrite the equation as:

$$\Delta Q = mc_v \Delta T$$

This is simply a linear relationship between ΔQ and ΔT, so we plot ΔQ against ΔT and find the slope, M. Then, it is a simple matter of dividing by mass to solve for c_v:

$$c_v = \frac{M}{m}$$

The graph we used, and the corresponding linear equation (with the slope used for calculation), for the first trial can be seen on the next page:

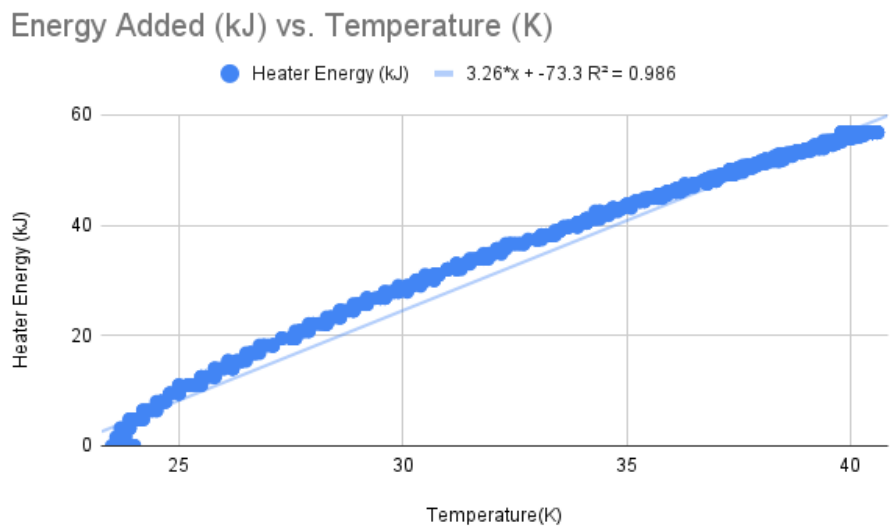


Figure 4: The energy added during the first trial plotted against the change in temperature. The slope from the linear fit was taken as the value of M.

The heat loss from the tank was determined by tracking the amount of energy input required to maintain a constant temperature. At this steady state, we have the following.

$$Q_{loss} = Q_{final} - Q_g$$

Where Q_{final} is simply the final value of the heat added. As we took the uncertainty in heat added to be 0.1 kJ, we simply take that value twice as our uncertainty in heat lost. For specific heat capacity, we propagate the uncertainty as follows. We take uncertainty in time to be 0.1 seconds. We propogate our uncertainty with the following formula, where T denotes the change in temperature for clarity.

$$\Delta c_v = c_v \sqrt{\left(\frac{\Delta Q_{\text{added}}}{Q_{\text{added}}}\right)^2 + \left(\frac{\Delta m}{m}\right)^2 + \left(\frac{\Delta(T)}{T}\right)^2}$$

The results are summarized in the table below:

Table 2: A table showing the initial temperature (T_i), final temperature (T_f), heat gained when ramping to the desired temperature (Q_g), heat lost due to conduction (Q_{loss}, and specific heat capacity at constant volume (c_v).

Trial #	T_i ($\pm$0.1K)	T_f ($\pm$0.1K)	Q_g ($\pm$0.1kJ)	Q_{loss} ($\pm$0.1kJ)	c_v ($\pm$0.1kJ/kg K)
1	297.0	313.05	56.8	20.6	95.9
2	296.95	313.25	39.2	49.1	96.8
3	298.45	332.55	107.7	80.3	118.3
4	301.15	333.15	113.5	120.1	62.4

Heat loss increased with both pressure and temperature, which is consistent with the theory that larger temperature gradients and increased pressure (therefore increased wall interactions) would increase heat loss. The average c_v value was roughly 93 kJ/(kg K), with an uncertainty larger than the value. The largest source of error would have been the fact that we considered the system to be just the air in the tank. The acrylic walls and tank top and bottoms would increase the amount of heat required to be added in order to maintain a constant temperature, which our equation did not account for.

4.3 Determining work done from propeller blades

We take the following information from the lab manual (CHE260, 2025a): The propeller has a diameter of 2.5 inches, or $D = 0.0635$ metres and a pump shaft speed of $n = 4200$ rpm at standard air conditions. Standard atmosphere is a reasonable approximation, as we performed the experiments at room temperature and pressure. We can relate the equations from (CHE260, 2025a), where P is the power done by the propeller:

$$\frac{Q_2}{Q_1} = \frac{n_2}{n_1}\left(\frac{D_2}{D_1}\right)^3$$
$$\frac{P_2}{P_1} = \left(\frac{n_2}{n_1}\right)^3\left(\frac{D_2}{D_1}\right)^5\left(\frac{\rho_2}{\rho_1}\right)$$

The fan diameter and rotational speed are virtually identical, as the reference conditions are close enough to actual experimental conditions. Thus, the expression for shaft power reduces to:

$$P_2 = P_1\left(\frac{\rho_2}{\rho_1}\right), \tag{1}$$

where ρ_2 is the air density in the tank and ρ_1 is the density of standard air. Standard air has density $\rho_1 = 1.27\frac{\text{kg}}{m^3}$ (National Institute of Standards and Technology (NIST), 2023). To calculate the density of

the air in the tank, we manipulate ideal gas law as follows.

$$\rho_{\text{tank}} = \frac{P}{RT}$$

Taking data from trial 4, we have $P = 642.5\text{kPa}$ and $T = 333.15\text{K}$. Using the gas constant for air $R = 287\text{J/kgK}$ (Smith et al., 2018), the density of the air in the tank is calculated as:

$$\rho_{\text{tank}} = \frac{642500}{287 \times 333.15} = 6.71\text{kg/m}^3$$

Using the provided reference power of $P_1 = 0.001\text{hp} = 0.746\text{W}$ (CHE260, 2025a), we can plug our values into the equation to find actual power done by the shaft:

$$\begin{aligned} P_2 &= 0.746\left(\frac{6.71}{1.27}\right) \\ &= 5.28\text{W} \end{aligned}$$

As we have a fixed volume, no work is done on the system. We can therefore rearrange the first law of thermodynamics as follows to balance the power input and the rate of heat rise, $\frac{dT}{dt}$ as follows:

$$\begin{aligned} P_2 &= \frac{dT}{dt} m c_v \\ \frac{dT}{dt} &= \frac{P_2}{m c_v} \end{aligned}$$

As our value of c_v is well in disagreement with the provided value (Smith et al., 2018), we will use the provided value of $c_v = 717\text{kJ/(kg K)}$ and our measured value of $m = 0.0579\text{kg}$.

$$\frac{dT}{dt} = \frac{5.28}{0.0579 \times 717} = 0.127\text{K/s}$$

As the heater provided approximately 1000 W of power, it is clear that the fan blades only providing roughly 5 W, or 0.5% of the power is a negligible source and can be safely ignored.

5 Conclusion

In this series of experiments, the first law of thermodynamics and the ideal gas law were used to model a tank of air system to determine the relationships between heat transfer, internal energy, and mechanical work. We determined the mass of the air by using the ideal gas law. The rate of heat loss from the tank

was obtained by determining the heat added at constant temperatures, and then by using the first law of thermodynamics. It was shown that the heat loss increased with both pressure and temperature, which is consistent with the theory that larger temperature gradients and increased pressure (therefore increased wall interactions) would increase heat loss. We also calculated the specific heat capacity of air at constant volume. The average value determined was 93 kJ/(kg K), with an uncertainty larger than the value. This is in significant disagreement with literature values, but technically within uncertainty. The largest source of error was hypothesized to be due to heat losses from the system, since we considered an adiabatic tank to calculate the value. Finally, the work done by the built in tank propeller was calculated with provided values and was shown to be roughly 0.5% of the total energy added to the system, which is negligible.

6 References

References

CHE260. (2025a). Che 260 - introductory manual [Department of Chemical Engineering, University of Toronto].

CHE260. (2025b). Che 260 lab 1 - ideal gas law [Department of Chemical Engineering, University of Toronto].

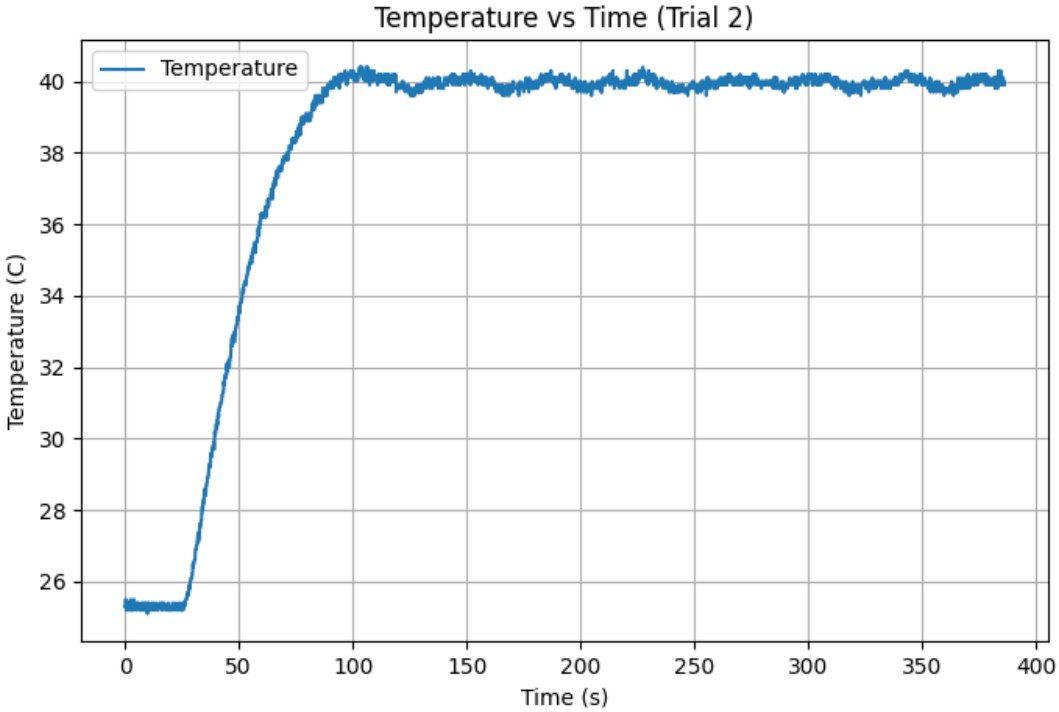
Howell, R. H. (2017). *Principles of heating, ventilating, and air conditioning: A textbook with design data based on the 2017 ashrae handbook–fundamentals* (8th ed.). ASHRAE.

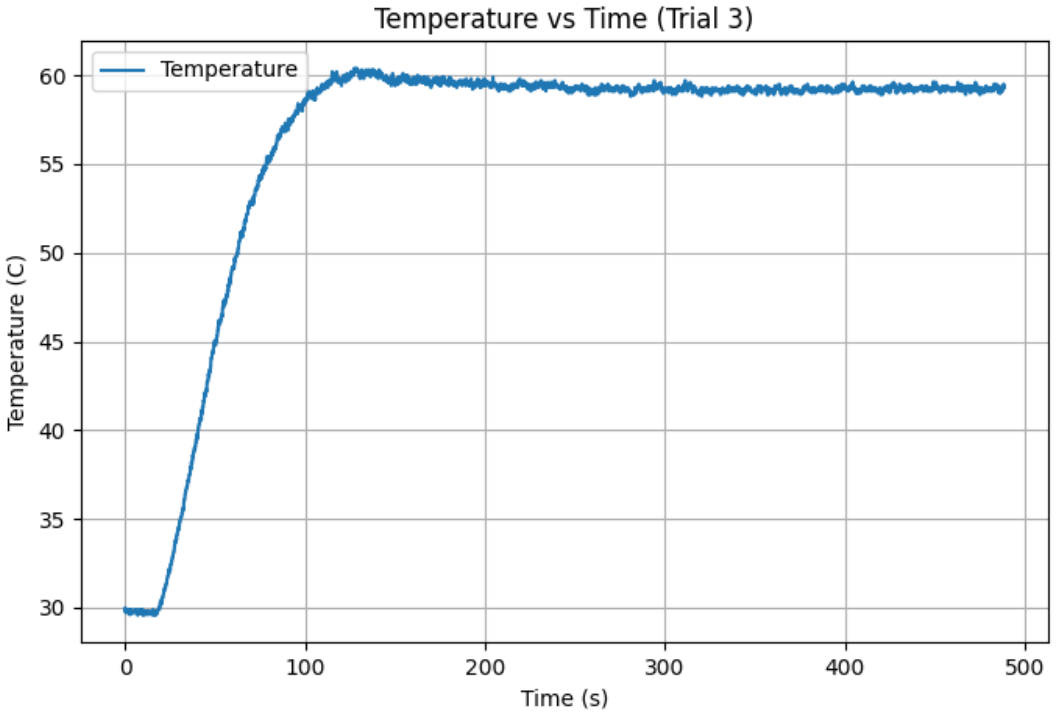
National Institute of Standards and Technology (NIST). (2023). Glossary term: Ideal gas law [Accessed: 2025-11-19].

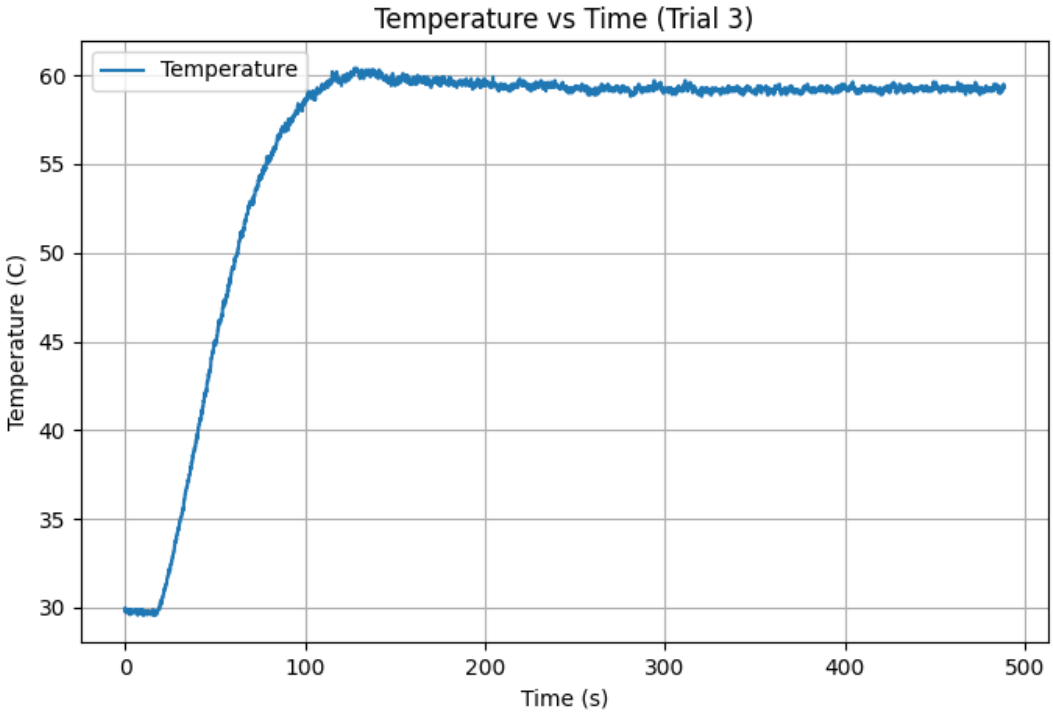
Smith, J. M., Ness, H. C. V., Abbott, M. M., & Swihart, M. T. (2018). *Introduction to chemical engineering thermodynamics* (8th ed.). McGraw-Hill Education.

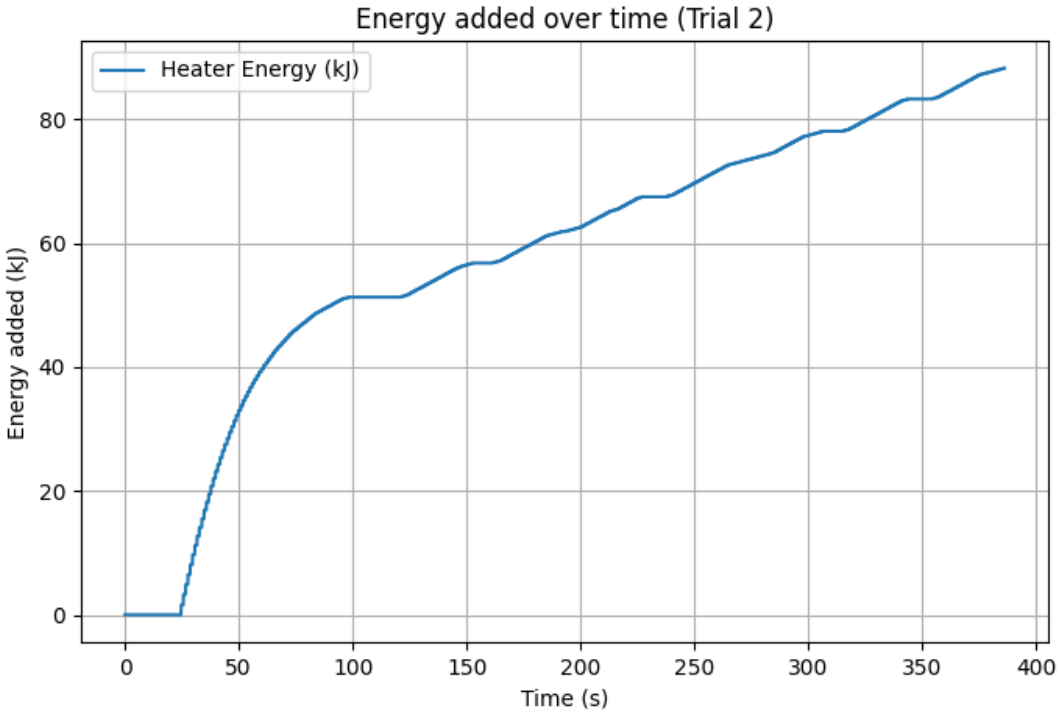
Sohn, H. Y., & Olivas-Martinez, M. (2014). Determination of energy requirements for ironmaking processes: It's not that straightforward. In *Energy technology 2014: Carbon dioxide management and other technologies* (pp. 49–56). Minerals, Metals; Materials Society. https://doi.org/10.1002/9781118888735.ch6

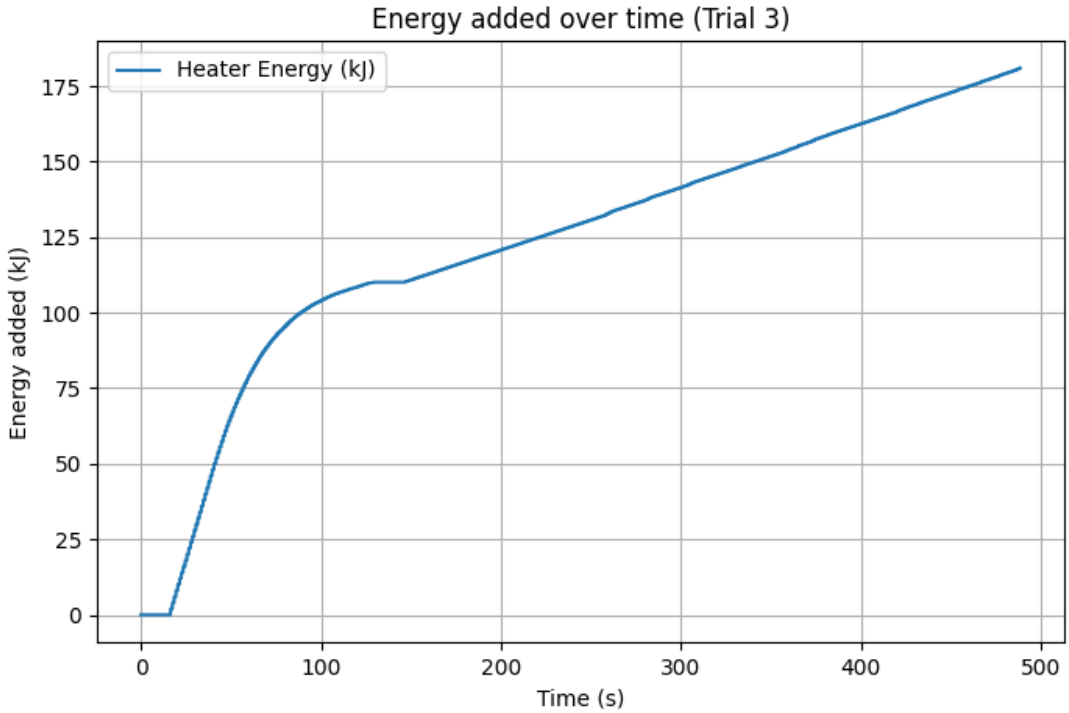
7 Appendix: additional graphs

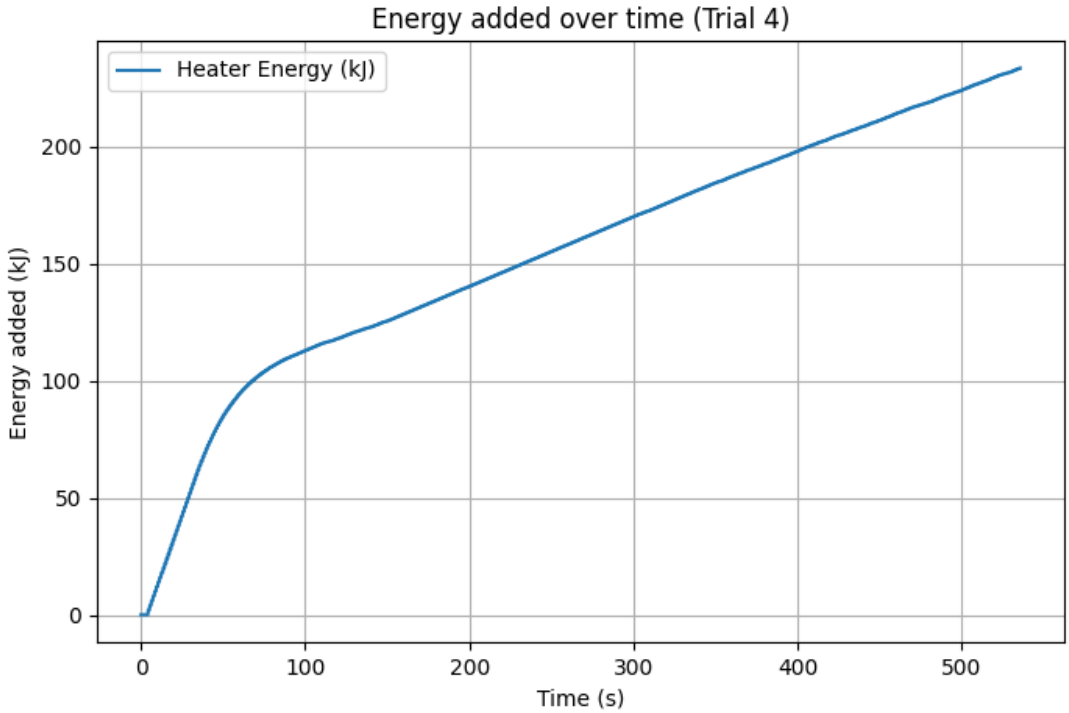





Temperature vs Time (Trial 3)
Temperature
Temperature (C)
Time (s)
60
55
50
45
40
35
30
0
100
200
300
400
500


Energy added over time (Trial 2)
Heater Energy (kJ)
Energy added (kJ)
Time (s)
80
60
40
20
0
0
50
100
150
200
250
300
350
400


Energy added over time (Trial 3)
Heater Energy (kJ)
Energy added (kJ)
Time (s)
0
25
50
75
100
125
150
175
0
100
200
300
400
500


Energy added over time (Trial 4)
Heater Energy (kJ)
Energy added (kJ)
Time (s)
0
50
100
150
200
0
100
200
300
400
500